Statistics Lab 52568 — Formula & Concept Sheet

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Coverage & Poisson Model

Coverage Y_i = number of reads starting at base i.

Uniform Poisson (3 assumptions):

1. Identical distribution: same p_i for every fragment
2. Uniformity: $p_i = 1/I$ for all i
3. Independence: fragments land independently

$\Rightarrow Y_i \sim \text{Poisson}(\lambda), \quad \lambda = J/I$

Key Poisson property: $\mathbb{E}[Y] = \text{Var}[Y] = \lambda$

$$\text{VMR} = \frac{\text{Var}[Y]}{\mathbb{E}[Y]} = 1 \quad \text{(if uniform)}$$

Binned coverage: bin k of size b: $Y_k = \sum_{\text{bases in } k} Y_i \sim \text{Poisson}(\lambda_k)$

Variance Decomposition (Law of Total Variance)

$$\text{Var}(Y) = \underbrace{\mathbb{E}[\text{Var}(Y \mid X)]}_{\text{avg within-group}} + \underbrace{\text{Var}(\mathbb{E}[Y \mid X])}_{\text{variance of group means}}$$

Applied to $Y_k \sim \text{Poisson}(\lambda_k)$:

$$\boxed{\text{Var}(Y_k) = \underbrace{\mathbb{E}[\lambda_k]}_{\text{Poisson noise}} + \underbrace{\text{Var}(\lambda_k)}_{\text{rate heterogeneity}}}$$

$$\text{VMR} = 1 + \frac{\text{Var}(\lambda_k)}{\bar{\lambda}}$$

VMR $= 1$ iff all λ_k equal (uniform model).
Spatial correlation of GC content prevents $\text{Var}(\lambda_k)$ from averaging out — hence VMR ≈ 14.7 in our data.

Regression for $f(\text{gc})$

Linear: $Y_k = \beta_0 + \beta_1 \, \text{gc}_k + \epsilon_k, \quad \hat{\beta}_1 = \widehat{\text{Cov}}(\text{gc}, Y)/\widehat{\text{Var}}(\text{gc})$

B-spline (cubic, knots $t_1, \dots, t_m$): Fit $Y_k \sim \text{bs}(\text{gc}_k, \text{knots}, \text{degree} = 3)$ via OLS. Ensures continuity of function and derivatives 1–2 at each knot.
More knots = more flexibility; degree controls smoothness.

Fit quality:

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}}, \quad \text{RMSE} = \sqrt{\frac{1}{K}\sum_k r_k^2}$$

Heteroscedasticity check:

$$\text{SD ratio} = \frac{\text{SD(residuals, gc} > \text{Q3)}}{\text{SD(residuals, gc} < \text{Q1)}}$$

SD ratio ≈ 2.25 in our data $\Rightarrow$ variance grows with coverage.

Poisson GLM & Quasi-Poisson

Poisson GLM: $\log \mathbb{E}[Y_k] = \beta_0 + \beta_1 \, \text{gc}_k$
Assumes $\text{Var}(Y) = \mathbb{E}[Y]$.
Quasi-Poisson: same mean, but $\text{Var}(Y) = \phi \, \mathbb{E}[Y]$, ϕ estimated.
In our data: $\phi \approx 14$ after GC is modelled.

McFadden pseudo-R^2: $\tilde{R}^2 = 1 - \dfrac{\log L(\hat{\text{model}})}{\log L(\text{null})}$

Copy Number Estimation

Model:

$$Y_k \sim \text{Pois}(\lambda_k), \quad \lambda_k = a_k \cdot f(\text{gc}_k) + \epsilon_k$$

Identifiability: $\text{med}_k\{a_k\} = 1$, $\mathbb{E}_k[f(\text{gc}_k)] = 1$, $\mathbb{E}[\epsilon_k] = 0$.

Plug-in estimator:

$$\hat{a}_k = \frac{y_k}{\hat{f}(\text{gc}_k)} \cdot \hat{M}, \quad \hat{M} = \frac{1}{\text{med}_k\left\{\dfrac{y_k}{\hat{f}(\text{gc}_k)}\right\}}$$

Variance decomposition of $\hat{a}_k$:

$$\mathrm{Var}(\hat{a}_k) \approx \frac{\lambda_k + \sigma^2}{f(\mathrm{gc}_k)^2} = \underbrace{\frac{1}{f(\mathrm{gc}_k)}}_{\text{Poisson}} + \underbrace{\frac{\sigma^2}{f(\mathrm{gc}_k)^2}}_{\varepsilon \text{ noise}}$$

In our data: Poisson $\approx 16\%$, $\varepsilon \approx 84\%$ of total variance.

Two-sample estimator:

$$\hat{a}_k^{\text{2-sample}} = \frac{\hat{a}_k^{\text{tumor}}}{\hat{a}_k^{\text{healthy}}}$$

Divides out shared GC bias and germline CNVs.

Noise models:

- Additive: $\lambda_k = a_k f + \epsilon$ (fixed-size offset)
- Multiplicative: $\lambda_k = a_k f \cdot \eta,\ \eta \sim N(1, \gamma^2)$ $\leftarrow$ more biologically plausible (error scales with signal)

Hypothesis Testing & Empirical Null

Statistic: $S1_k = \log_2(\hat{a}_k^{\text{2-sample}})$
Under H_0 (no tumor-specific change): $S1 \approx 0$.

Negative control selection (Strategy B):
Keep bin k if in its 11-bin window: $|\tilde{T}| \leq \log_2 1.10$ AND SD $\leq \log_2 1.15$.
Filter on the *ratio*, not the healthy alone: H_0 is "no tumor-specific change", so shared germline CNVs are correctly included.

Empirical two-sided p-value:

$$p_k = \frac{\big|\{j \in \mathrm{ctrl} : |S1_j| \geq |S1_k|\}\big|}{|\mathrm{ctrl}|}$$

Requires no distributional assumption; resolution floor $= 1/|\mathrm{ctrl}|$.

Calibration check: split control bins into interleaved blocks $\gg W$; compute p-values of held-out CN=1 bins; check P-P plot tracks the uniform diagonal.
$\Rightarrow$ KS $D \approx 0.009$ (S1 well-calibrated).

Multiple Testing Correction

Bonferroni (FWER control): $\alpha' = \alpha/m$.
Probability of ≥ 1 false positive $\leq \alpha$.
Too strict for genome-wide scan ($m \sim 10^5$).

Benjamini–Hochberg (FDR control):
Sort $p_{(1)} \leq \cdots \leq p_{(m)}$. Reject all $H_{(k)}$ with

$$k \leq k^* = \max\left\{k : p_{(k)} \leq \frac{k}{m}\alpha\right\}.$$

Controls $\mathbb{E}\left[\frac{\text{false positives}}{\text{total significant}}\right] \leq \alpha$.
Preferred for genome-wide tests: tolerates a known small fraction of false positives in exchange for power to detect many real events.

Bias–Variance Tradeoff / Bin Size

Larger bins $\Rightarrow \downarrow$ per-bin Poisson noise, $\uparrow$ GC signal blurring.
Statistical elbow: $\sim$5 kb (marginal R^2 gain $< 5\%$ threshold).
Biological constraint: genes 10–100 kb, CpG islands 0.5–2 kb.
Aggregative model: fit at fine resolution, predict parent cells by summing sub-cell predictions — best of both (1 kb wins in lab 7).

Key Numbers

λ (50–60 Mb)	≈ 0.036 reads/base
Binned VMR	≈ 14.7
GC–coverage r	≈ 0.84
Spline R^2 ceiling	≈ 0.75
SD ratio	≈ 2.25
Quasi-Poisson ϕ	≈ 14
CN MAE (healthy)	≈ 0.15
Var split (Pois/ε)	16% / 84%
r (tumor vs healthy)	≈ 0.10
FDR gains / losses	1600 / 1313
Focal deletion	$\sim$28.5 Mb, ratio ≈ 0.54